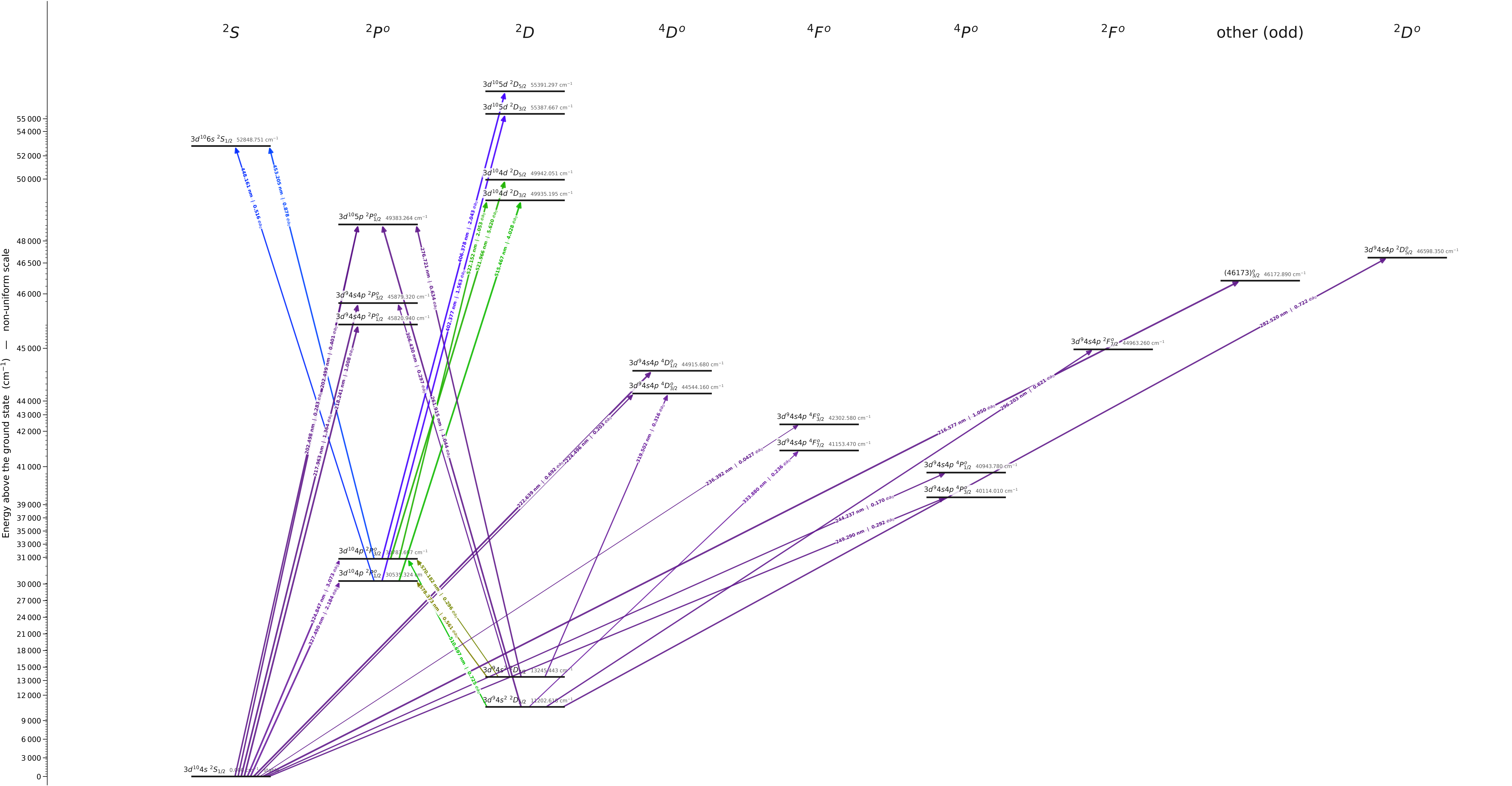
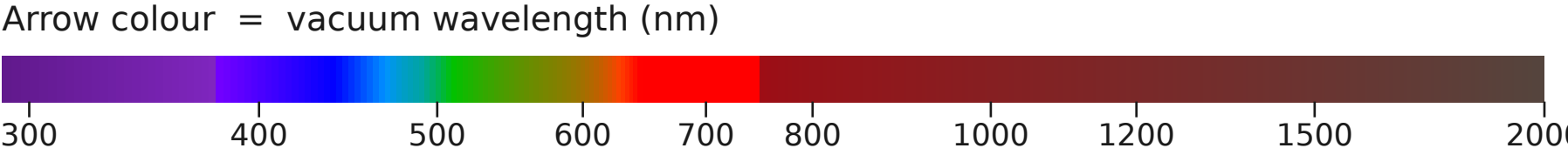


Copper

Cu (Cu I). Strongest classified transitions below 2 μm (NIST): 23 levels, 29 lines.

Cu⁺ ionisation limit 62317.46 cm⁻¹ = 7.72638 eV (160.469 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[U\|er\|J']$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$[U\ er\ J']$ (ea_0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$3d^{10}4s\ ^2S_{1/2} - 3d^{10}5p\ ^2P_{1/2}^o$	202.4978	202.4326	1480.4730	0.283	9.800e+06	-	-	
$3d^{10}4s\ ^2S_{1/2} - 3d^{10}5p\ ^2P_{3/2}^o$	202.4991	202.4339	1480.4634	0.401	9.800e+06	-	-	
$3d^{10}4s\ ^2S_{1/2} - (46173)\frac{3}{2}$	216.5773	216.5093	1384.2284	1.050	5.500e+07	-	-	
$3d^{10}4s\ ^2S_{1/2} - 3d^94s4p\ ^2P_{3/2}^o$	217.9631	217.8949	1375.4274	1.364	9.100e+07	-	-	
$3d^{10}4s\ ^2S_{1/2} - 3d^94s4p\ ^2P_{1/2}^o$	218.2408	218.1725	1373.6772	1.008	9.900e+07	-	-	
$3d^{10}4s\ ^2S_{1/2} - 3d^94s4p\ ^4D_{3/2}^o$	222.6394	222.5702	1346.5382	0.692	4.400e+07	-	-	
$3d^94s^2\ ^2D_{5/2} - 3d^{10}5p\ ^2P_{3/2}^o$	261.9150	261.8368	1144.6174	1.044	3.070e+07	-	-	
$3d^94s^2\ ^2D_{3/2} - 3d^{10}5p\ ^2P_{3/2}^o$	276.7208	276.6391	1083.3751	0.634	9.600e+06	-	-	
$3d^{10}4s\ ^2S_{1/2} - 3d^{10}4p\ ^2P_{3/2}^o$	324.8473	324.7536	922.8720	3.073	1.395e+08	-	-	
$3d^{10}4s\ ^2S_{1/2} - 3d^{10}4p\ ^2P_{1/2}^o$	327.4896	327.3952	915.4260	2.184	1.376e+08	-	-	
$3d^{10}4p\ ^2P_{1/2}^o - 3d^{10}5d\ ^2D_{3/2}$	402.3765	402.2629	745.0545	1.563	1.900e+07	-	-	
$3d^{10}4p\ ^2P_{3/2}^o - 3d^{10}5d\ ^2D_{5/2}$	406.3785	406.2638	737.7173	2.043	2.100e+07	-	-	
$3d^{10}4p\ ^2P_{3/2}^o - 3d^{10}6s\ ^2S_{1/2}$	453.2053	453.0783	661.4937	0.878	8.400e+06	-	-	
$3d^{10}4p\ ^2P_{1/2}^o - 3d^{10}4d\ ^2D_{3/2}$	515.4673	515.3238	581.5935	4.028	6.000e+07	-	-	
$3d^{10}4p\ ^2P_{3/2}^o - 3d^{10}4d\ ^2D_{5/2}$	521.9655	521.8202	574.3530	5.620	7.500e+07	-	-	
$3d^{10}4p\ ^2P_{3/2}^o - 3d^{10}4d\ ^2D_{3/2}$	522.1524	522.0070	574.1475	2.053	1.500e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).